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GENERATIVE AI MINI PROJECT REPORT

On

“Cold Email Generator”

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in

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ABSTRACT

Customer churn remains a critical challenge for the telecommunications industry, directly impacting revenue and long-term growth. This project presents an end-to-end Machine Learning pipeline and interactive web dashboard designed to predict, explain, and mitigate customer attrition. At its core, the system utilizes a high-accuracy XGBoost classifier trained on comprehensive customer data—including demographics, service usage, and billing patterns—to accurately identify individuals at risk of canceling their subscriptions.

To ensure transparency and trust in the model's decisions, the system integrates Explainable AI (XAI) using SHAP (SHapley Additive exPlanations). This allows the platform to break down the specific factors driving each customer's churn probability, answering exactly *why* a customer is at risk. Beyond mere prediction, the system proactively generates targeted, cost-effective retention strategies and calculates the expected Return on Investment (ROI) for intervening.

The solution is wrapped in a responsive, real-time Flask-based web application that allows stakeholders to monitor churn metrics, visualize feature contributions, and perform bulk predictions via Excel uploads. By bridging the gap between predictive modeling and actionable business insights, this project provides a comprehensive, scalable tool for telecom providers to proactively manage customer relationships and reduce revenue leakage.

Keywords: Customer Churn Prediction, Machine Learning, XGBoost, Explainable AI (XAI), SHAP, Predictive Analytics, Telecommunications, Retention Strategies, Return on Investment (ROI), Web Dashboard.

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Chapter 1

INTRODUCTION

1.1 Problem Statement

In the highly competitive telecommunications industry, customer churn causes significant revenue leakage. Because acquiring new customers is far more expensive than retaining existing ones, proactive churn management is essential. However, telecom providers struggle with two main challenges: traditional systems cannot accurately predict which specific customers will leave, and modern Machine Learning models often act as "black boxes" that fail to explain *why* a prediction was made.

There is a critical need for an intelligent system that not only accurately predicts customer churn but also provides transparent explanations for its decisions. This project addresses this gap by creating an explainable Machine Learning pipeline that identifies at-risk customers, explains the underlying factors driving their dissatisfaction, and recommends actionable, cost-effective retention strategies to minimize revenue loss.

1.2 Objectives

- Develop a Predictive Churn Model
- Implement Explainable AI (XAI)
- Generate Actionable Retention Strategies
- Calculate Intervention ROI
- Build an Interactive Web Dashboard
- Enable Bulk Data Processing

1.3 Scope

The scope of Cold Email Generator includes:

- Tabular Data Processing
- Binary Classification Prediction
- Feature-Level Explainability
- Automated Rule-Based Retention
- Local/Containerized Deployment
- Batch and Real-Time Inference

1.4 Methodology

The project follows a modular architecture combining frontend development, edge-native backend orchestration, vector retrieval systems, and Generative AI workflows.

The methodology includes:

- Data Collection & Generation
- Exploratory Data Analysis (EDA)
- Data Preprocessing & Feature Engineering
- Model Selection & Training (XGBoost)
- Model Evaluation & Validation
- Explainable AI Integration (SHAP)
- Web Application & API Deployment

Chapter 2

LITERATURE SURVEY

2.1 Existing Systems

Existing operational simulation systems primarily rely on static workflows and predefined rule engines. Traditional crisis-management systems provide limited contextual reasoning and lack intelligent retrieval capabilities.

Most conventional systems suffer from:

- Limited semantic understanding
- No Retrieval-Augmented Generation support
- Poor protocol retrieval accuracy
- Lack of operational memory systems
- Limited simulation adaptability

2.2 Proposed Solution

Scenario Weaver introduces a lightweight edge-native Generative AI platform integrating:

- Retrieval-Augmented Generation (RAG)
- pgvector-based semantic search
- Hybrid retrieval pipelines
- Event-driven simulation workflows
- Agent memory systems
- Automated evaluation and AAR generation

The proposed system improves contextual reasoning, operational retrieval accuracy, and simulation realism.

Chapter 3

REQUIREMENT ANALYSIS AND SPECIFICATION

3.1 Functional Requirements

Sl.No	Requirement Description
1	The system must allow users to upload PDF and TXT operational documents.
2	The system must perform semantic retrieval using pgvector.
3	The system must support hybrid retrieval pipelines.
4	The system must simulate operational events dynamically.
5	The system must maintain agent memory and event history.
6	The system must generate evaluation reports and AARs.
7	The system must support retrieval debugging and protocol inspection.
8	The system must support scalable edge-native deployment.

3.2 Software Requirements

React

TypeScript

Tailwind CSS

Supabase

PostgreSQL

pgvector

Deno Runtime

Node.js

VS Code

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

Scenario Weaver follows an edge-native Generative AI architecture designed for operational simulations, protocol intelligence, and AI-driven forecasting workflows. The platform integrates Retrieval-Augmented Generation (RAG), hybrid semantic retrieval, event-driven orchestration, and relationship graph modeling into a unified simulation environment.

The system enables users to:

- Create AI-powered operational simulations
- Upload and retrieve operational protocols
- Generate relationship graphs between entities
- Simulate multi-agent interactions
- Perform event-driven forecasting
- Generate evaluations and After-Action Reports

4.2 Purpose of the System

The primary purpose of Scenario Weaver is to provide an intelligent operational reasoning platform capable of simulating crisis-response workflows, forecasting operational outcomes, and analyzing relationships between agents, systems, and hazards.

The system combines Generative AI and hybrid retrieval pipelines to improve contextual reasoning and decision-support capabilities.

4.3 System Modules

4.3.1 Simulation Engine

The simulation engine dynamically generates operational scenarios using Generative AI models. It supports:

- Multi-agent simulations

- Event progression

- Scenario branching

- Forecast generation

- Escalation modeling

4.3.2 Protocol Intelligence Module

This module allows users to upload operational documents such as SOPs, incident manuals, hazard guidelines, and emergency-response protocols.

The system performs:

- Document parsing

- Text chunking

- Embedding generation

- Semantic indexing

- Hybrid retrieval

4.3.3 Relationship Graph Module

The relationship graph module models dependencies and interactions between operational entities, institutions, actors, hazards, and events.

The graph supports:

Influence analysis

Dependency tracking

Contagion modeling

Escalation analysis

Causal reasoning

4.3.4 Evaluation Module

The evaluation engine analyzes generated simulations and produces:

Operational assessments

Decision evaluations

Failure summaries

Timeline reconstruction

After-Action Reports (AAR)

4.4 Workflow of the System

- Data Acquisition and Loading
- Data Cleaning and Preprocessing
- Feature Encoding and Transformation
- Data Splitting (Train/Test Sets)
- XGBoost Model Initialization and Training
- Performance Metric Calculation
- SHAP Explainer Initialization
- Prediction and Explainability Generation

- Rule-Based ROI and Retention Calculation
- Data Persistence and Dashboard Visualization

4.5 Advantages of the System

- High Predictive Accuracy
- Transparent Decision Making
- Actionable Business Insights
- Financial Accountability
- Scalable Batch Processing
- User-Friendly Monitoring
- Proactive Revenue Protection

4.6 System Architecture Layers

4.6.1 Presentation Layer

Built using:

React

Vite

TypeScript

Tailwind CSS

This layer handles:

Simulation interfaces

Protocol uploads

Graph visualization

Dashboard analytics

Retrieval debugging

4.6.2 Application Layer

The application layer is implemented using Supabase Edge Functions and Deno runtime.

It performs:

Hybrid retrieval

Simulation orchestration

Event management

Relationship graph generation

Evaluation processing

Hybrid Retrieval Formula:

$$\text{Final Score} = (0.7 \times \text{Vector Similarity}) + (0.3 \times \text{Keyword Rank}) + \text{Identifier Boost}$$

4.6.3 Data Layer

The data layer uses PostgreSQL and pgvector for:

Vector storage

Semantic retrieval

Event persistence

Agent memory

Simulation history

Figure 4.1: THE LAND PAGE

The landing page provides an interactive interface for creating AI-driven operational simulations. Users can enter custom scenarios or select predefined templates such as banking crises, geopolitical conflicts, climate events, and supply-chain disruptions. The interface demonstrates the platform's ability to support multi-domain operational forecasting and Generative AI-based scenario modeling.

Figure 4.2: CREATION OF THE EMAIL

The simulation dashboard displays AI-generated operational scenarios, agent pipelines, retrieval workflows, and cited knowledge sources. The system performs input analysis, protocol retrieval, actor identification, and event orchestration to generate context-aware operational forecasts. The dashboard also supports multi-horizon simulation reasoning and source-grounded AI outputs.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Implementation

Scenario Weaver is implemented as a Generative AI-powered web platform.

The frontend uses React and TypeScript for responsive UI development. Supabase Edge Functions provide lightweight backend orchestration. PostgreSQL and pgvector are used for semantic vector retrieval.

The document ingestion pipeline performs:

1. PDF/TXT parsing
2. Text extraction
3. Chunk generation
4. Embedding creation
5. Vector indexing
6. Metadata storage

The simulation engine generates AI-driven operational workflows using event-based orchestration.

5.2 Testing

5.2.1 Functional Testing

The following modules were tested:

- Protocol uploads
- Hybrid retrieval
- Simulation execution
- Event generation
- Evaluation pipelines
- AAR generation

5.2.2 Performance Testing

Performance testing was conducted for:

- Retrieval latency

- Vector search performance

- Simulation response time

- Dashboard rendering

5.3 Results

The platform successfully:

- Retrieves operational protocols accurately

- Generates AI-driven simulations

- Maintains event memory

- Supports hazard-code reranking

- Produces automated evaluations and AARs

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

Scenario Weaver successfully demonstrates the integration of Generative AI, Retrieval-Augmented Generation, hybrid retrieval systems, and event-driven orchestration into a unified operational simulation platform.

The project provides protocol-aware operational reasoning, scalable retrieval systems, AI-generated simulations, and automated evaluation pipelines while maintaining lightweight edge-native deployment architecture.

The platform highlights the practical application of Generative AI in operational intelligence, crisis-response modeling, and simulation-based decision support systems.

6.2 Future Enhancements

Future improvements include:

- Real-time simulation streaming
- Multi-agent orchestration
- Knowledge graph visualization
- Temporal event analytics
- Voice-assisted simulations
- Advanced reranking models
- Multimodal document understanding